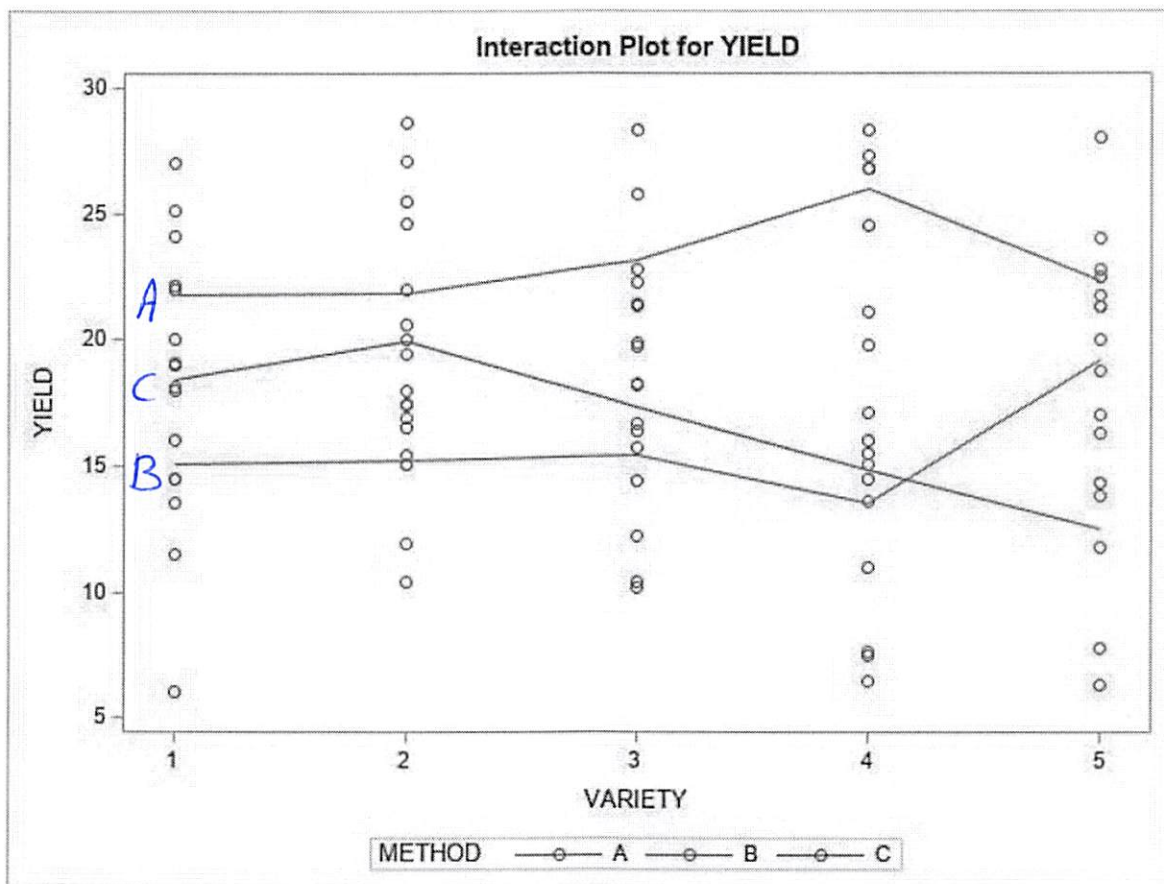


Multiple comparisons for grass data

Source	DF	Type III SS	Mean Square	F Value	Pr > F
METHOD	2	953.1562222	476.5781111	24.25	<.0001
VARIETY	4	11.3804444	2.8451111	0.14	0.9648
METHOD*VARIETY	8	374.4882222	46.8110278	2.38	0.0241



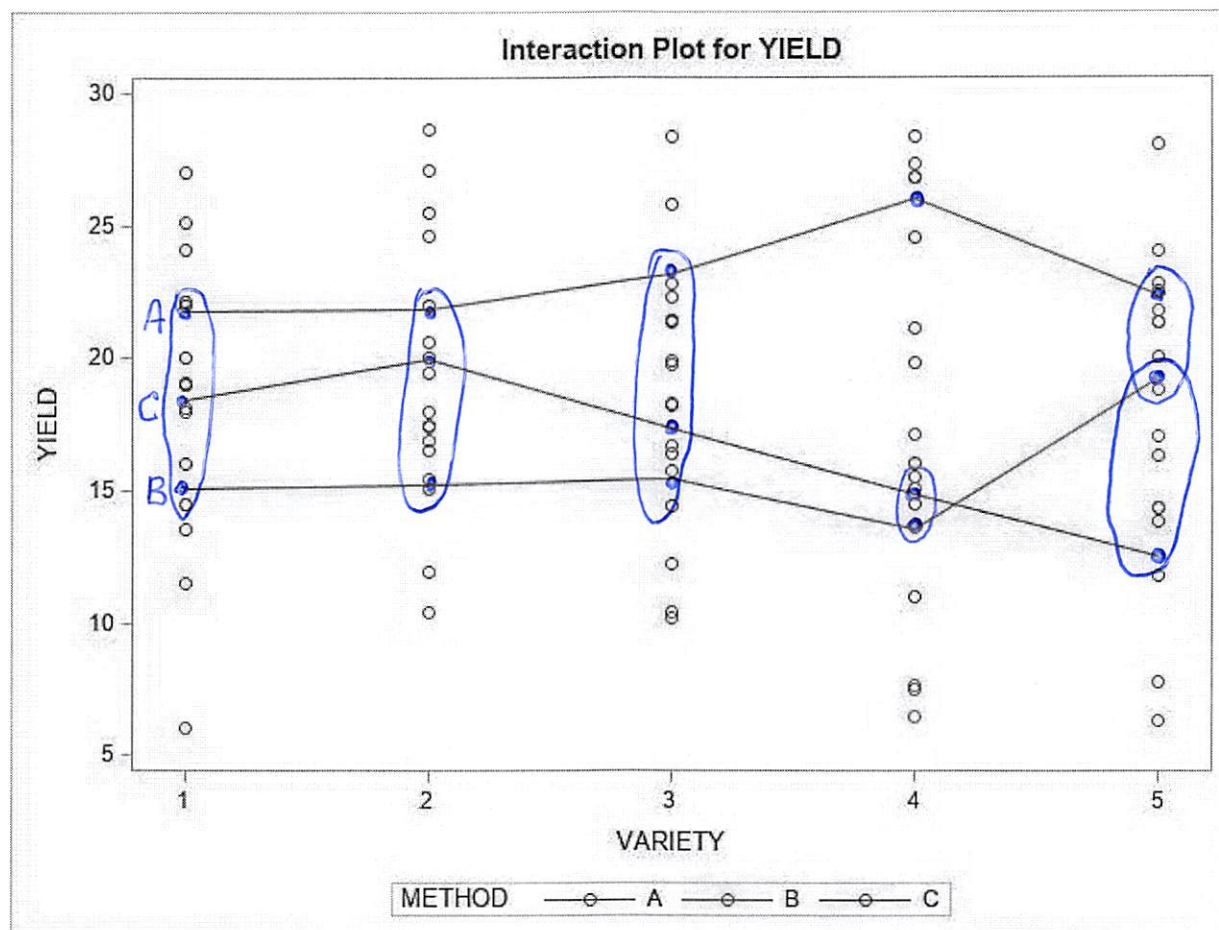
In order to compare simultaneously the three methods for each of the five varieties of grass, we need to calculate the Tukey's "yardstick". For a significance level of $\alpha = .05$, we get:

$$\begin{aligned}
 T_{.05} &= q_{.05}(a, f) \sqrt{\frac{MS_E}{n}} = q_{.05}(15, 75) \sqrt{\frac{19.65}{6}} \\
 &= 5.0 \sqrt{\frac{19.65}{6}} = 9.05
 \end{aligned}$$

Therefore, for each of the five varieties we will compare the means for the three methods. Any pair of means that differ by 9.05 units or more will be declared significantly different from each other.

Here are the sample means for methods for each variety:

Level of VARIETY	Level of METHOD	N	YIELD	
			Mean	Std Dev
1	A	6	21.7666667	2.73252020
1	B	6	15.0833333	7.05277723
1	C	6	18.4166667	2.72794184
2	A	6	21.8500000	5.88854821
2	B	6	15.2333333	3.44480285
2	C	6	19.9166667	3.36773910
3	A	6	23.1333333	3.50238014
3	B	6	15.4500000	3.61593695
3	C	6	17.3166667	4.40927054
4	A	6	25.9666667	3.07679487
4	B	6	13.5000000	5.60535458
4	C	6	14.8333333	5.72421756
5	A	6	22.3333333	3.71034590
5	B	6	19.2166667	3.36773910
5	C	6	12.5500000	5.35490429



Based on the multiple comparisons marked on the interaction plot of Method vs Variety, one can conclude that for varieties 1,2 and 3 methods A, B, C have not performed significantly different from each other.

For variety no 4, Method A performed significantly better than Method B and C. Methods B and C have not performed significantly different from each other.

For variety no 5, Method A performed significantly better than Method C. The other two pairwise comparisons have not shown significant differences.